Same Parts, Different Arrangements Matched Edits for Budgeted Anomaly Inspection

Anonymous CVPR submission

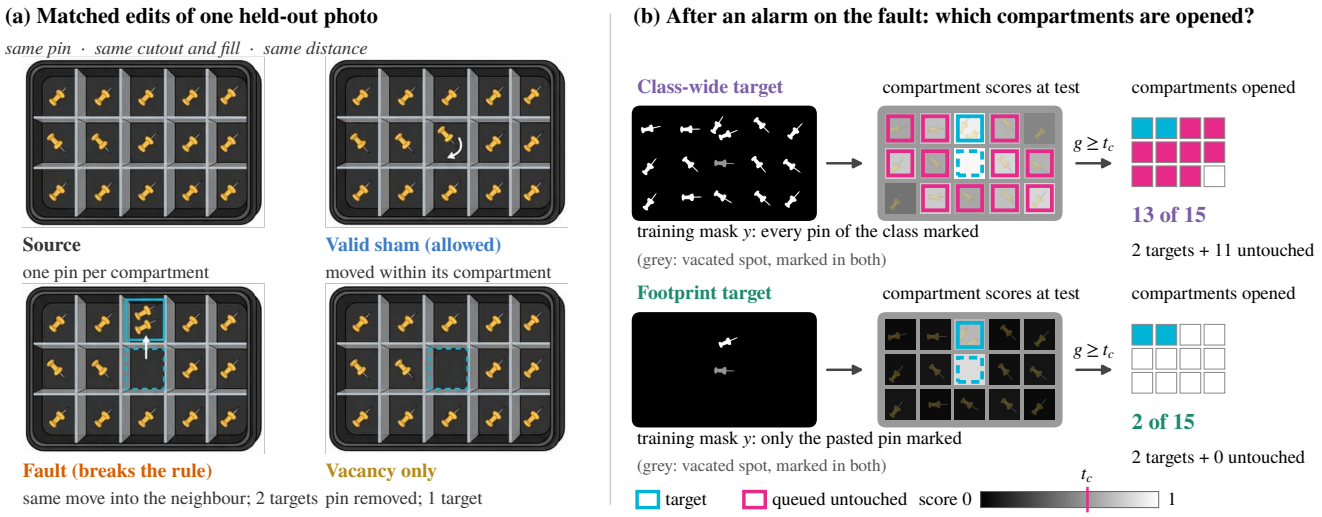


Figure 1. **Where should an inspector look after an alarm?** (a) Matched edits of one tray (rule: one pin per compartment): the same pin moves the same distance within its compartment (valid, no target) or into an occupied neighbour (fault). Cyan: compartments to check (dashed: vacated; solid: recipient). (b) Schematic: marking every pin of the inserted class in training makes nearly the whole tray score high, so 13 of 15 compartments are opened; marking only the pasted pin opens just the 2 targets. Measured at 95% recall: 11.07 vs. 0.20–0.45 untouched compartments (Tab. 1). Tray art created in BioRender.

Abstract

Anomaly detectors for industrial inspection are evaluated by whether they flag a defective product, yet an alarm is useful only if it also tells the inspector where to look. We study this gap for logical anomalies, where every part is intact but the arrangement violates a rule. Starting from 138 held-out photographs of a pushpin tray that must hold one pin per compartment, we construct matched edits in which the same pin is displaced by the same distance either within its compartment, which is valid, or into an occupied neighbour, which is a fault. The edits fix the two compartments that require inspection and allow us to measure, at a cutoff calibrated on development data, how many untouched compartments a detector queues. The released SALAD composition branch detects every fault, but at 95% target recall it queues 11.07 of the 13 untouched compartments per fault. Fine-tuning only this branch with targets that mark the pasted part, rather than every part of the inserted class, reduces the cost to 0.20 and 0.45 compartments in two seeds while every fault is still detected; one seed, however, raises more false alarms on valid edits. The same design identifies which SALAD branch separates a valid from an invalid edit under a second rule, and exposes a degree pathway in a masked graph model.

1. Introduction

Visual anomaly detection is usually assessed by whether a system flags defective products [3, 4, 16]. On a production line, however, an alarm is only the first step: an operator must then locate the defect. For *logical* anomalies [5] this second step is particularly demanding. A pushpin tray with one empty and one doubled compartment contains the correct number of intact pins, and only their arrangement is wrong. A detector that flags such a tray is helpful if its map points to the two affected compartments, and of little use if

it highlights the remaining thirteen as well.

Current benchmarks do not measure this cost. MVTec LOCO [5] reports image-level AUROC for detection and saturated per-region overlap for localization, and neither quantity states how many locations an inspector must examine to recover nearly every problem. Measuring it requires knowing exactly which locations are wrong, which natural logical defects rarely provide; an additional screw in a bag, for instance, may be any of several identical screws [6].

We obtain exact targets by intervention (Fig. 1a). From each held-out normal photograph we cut out an intact pin and move it either within its compartment, producing a *valid sham*, or by the same distance into an occupied neighbour, producing a *fault*; a third variant only removes the pin. Because all variants share the cutout, the background fill and the displacement, any difference in the detector's response is attributable to the rule violation, and the fault determines the two compartments that must be inspected. On top of these edits we define an inspection decision with an explicit cost (Sec. 3): a scene alarm decides whether a tray is opened, and a compartment cutoff fixed on development data decides which compartments are queued. We report the fraction of target compartments found (recall) together with the number of untouched compartments queued (cost).

We apply this protocol to SALAD [6], a recent logical anomaly detector whose composition branch scores predicted part maps, and ask two questions: how much inspection effort does a detector that catches every fault still impose, and what in its training determines that effort? Our paper offers the following contributions. As an **evaluation design**, we introduce matched valid and invalid edits with identical pixel changes, together with a recall–cost criterion calibrated on development data (Sec. 3). As a **diagnosis**, we show that the released composition branch detects all 138 faults and ranks each above its sham, yet queues 11.07 of 13 untouched compartments per fault at 95% target recall (Sec. 4.1). As a **remedy**, we show that the training target governs this cost: marking only the pasted part reduces it to 0.20–0.45 compartments across two seeds, with paired intervals far from zero (Sec. 4.2). Finally, as **transfer checks**, the same design identifies which SALAD branch carries the evidence under a second rule and isolates a degree pathway in a masked link predictor (Secs. 5 and 6).

2. Related work

Industrial anomaly detection was first benchmarked on local structural defects such as scratches and dents [4], where patch-feature memory banks [16] and student–teacher networks [3] perform strongly. MVTec LOCO [5] adds logical defects that violate constraints on the number, position or combination of parts and evaluates localization with the saturated per-region overlap (sPRO), which tolerates ambiguous ground truth. Component-based methods segment parts and compare attributes, correspondences or executable rules [9, 13, 15, 21], and SALAD [6] combines a composition branch on part maps with local and global appearance branches. Many of these detectors train a discriminative head on synthetic defects: CutPaste [11] pastes image patches, DRAEM [19] blends Perlin-masked textures and segments the altered region, LogicAL [22] synthesizes logical defects, and SALAD inserts and relabels parts in composition maps. This line of work establishes strong detection and region overlap, but it neither measures how many locations an inspector must open once an alarm fires nor examines how the target mask of the synthetic edits shapes that number. We hold the detector and its edits fixed and study exactly these two questions.

Our evaluation borrows from matched and counterfactual testing outside anomaly detection. Contrast sets [7] and counterfactually augmented data [10] probe language models with minimally different inputs that carry different labels, and Winoground [18] pairs captions that share words but differ in their relations; such pairs reveal shortcut features that ordinary test sets reward [8]. Our valid shams play this role for anomaly maps, since they alter as many pixels as the fault while respecting the rule. Operating points follow split-conformal calibration on normal photographs [1, 2], which bounds the false-alarm rate, and for graphs we build on the known effects of target-link masking and node degree in link prediction [20, 23], measuring how masking changes the available signal when two graphs share the same degree sequence.

3. Method

3.1. Problem setting

A photograph x_i shows a tray of $|\mathcal{E}| = 15$ compartments arranged in a 5×3 grid, and the published rule requires exactly one pin per compartment [5]. A part extractor maps the photograph to a six-class part map $m_i \in \{0,\dots,5\}^{256\times256}$, and the composition branch f of the detector returns a score map $S_i = f(m_i) \in [0,1]^{256\times256}$ in which higher values indicate anomalies (Fig. 2). For a compartment $e \in \mathcal{E}$, $S_i|_e$ denotes the scores of its pixels; compartment boundaries are registered once from the released label maps.

3.2. Matched edits

We use all 138 held-out normal pushpin photographs. Using the released class mask and a two-pixel RGB halo, we cut out one intact pin and translate it without resampling, which yields four variants per photograph (Fig. 1a, Fig. 3). The *source* is the unedited photograph. In the *valid sham*, the pin moves within its own compartment, so every compartment still holds one pin. In the *fault*, the pin moves by the same squared displacement into a face-adjacent occupied compartment, leaving its source compartment a_i empty and the

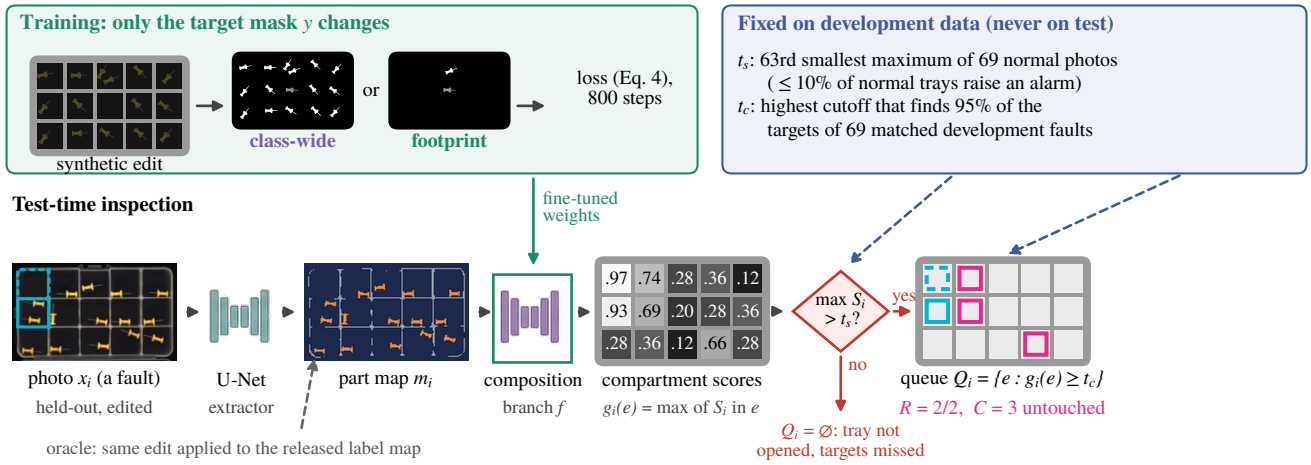


Figure 2. **Only the training target changes; everything at test time is fixed in advance.** Training (green) differs only in the target mask. At test, a part map comes from a U-Net (*extracted*) or from the edited label map (*oracle*); the scene alarm (t_s) and the compartment cutoff (t_c) are fixed on development data (blue). The values are illustrative and match the example under Eq. (3) ($R = 2/2$, $C = 3$). Network icons created in BioRender.

recipient b_i with two pins. The *vacancy-only* variant removes the pin without pasting it elsewhere. The sham and the fault share the cutout, the fill (the median of the local dark background) and the paste area, and therefore change exactly the same number of RGB pixels; collisions, divider clearance and background agreement are verified before any score is computed. Displacements range from 19 to 51 pixels.

Each variant has a target set $\mathcal{T}_i$ of compartments that require inspection: $\mathcal{T}_i = \{a_i, b_i\}$ for a fault, $\mathcal{T}_i = \{a_i\}$ for a vacancy, and $\mathcal{T}_i = \emptyset$ for a source or a sham, where every queued compartment is a false mark. Since the vacancy-only edit retains the empty half of the fault, comparing the two isolates the effect of the additional pin.

Every variant is scored through two interfaces with identical detector weights. The *oracle* interface moves the pin directly in the released label map, while the *extracted* interface re-segments every RGB variant, including the source, with an independent U-Net trained only on normal training photographs (97.3% development pin IoU). Since a deployed system never observes ground-truth labels, we report the extracted interface unless stated otherwise.

3.3. Calibrated inspection queue

We model the two decisions of an inspector, whether to open a tray and which compartments to check, with cutoffs that are fixed before testing (Algorithm 1). A tray raises an alarm when its highest pixel score exceeds a scene cutoff t_s,

$$\text{alarm}_i = \mathbb{1}\big[\max_p S_i(p) > t_s\big], \tag{1}$$

where t_s is the 63rd smallest maximum among 69 unedited validation photographs. This strict split-conformal cutoff [2] ensures that at most 10% of normal trays raise an alarm. Each compartment then receives its maximum score $g_i(e) = \max S_i|_e$, and the inspection queue at compartment cutoff t_c is

$$\mathcal{Q}_i = \begin{cases} \{e \in \mathcal{E} : \ g_i(e) \geq t_c\} & \text{if alarm}_i = 1, \\ \emptyset & \text{otherwise,} \end{cases} \tag{2}$$

so that the targets of a tray without an alarm count as missed. Over n faults, the target recall R and the inspection cost C are

$$R = \frac{1}{2n}\sum_{i=1}^{n} |\mathcal{Q}_i \cap \mathcal{T}_i|, \qquad C = \frac{1}{n}\sum_{i=1}^{n} |\mathcal{Q}_i \setminus \mathcal{T}_i|, \tag{3}$$

i.e., the fraction of target compartments that are queued and the mean number of untouched compartments queued per fault, which ranges from 0 to 13. For example, a fault whose queue contains both targets and three other compartments contributes $2/2$ to recall and 3 to cost.

The compartment cutoff is fitted on 69 matched *development* faults, constructed in the same way from validation photographs, and never on test data. The *95% cutoff* is the highest threshold that attains 95% target recall on these faults; with $2 \times 69 = 138$ development targets it is the 132nd largest target score, since $\lceil 0.95 \cdot 138 \rceil = 132$. A *fixed-cost cutoff* instead admits at most 0.5 untouched compartments per development fault and is the 35th largest untouched score, since $\lfloor 0.5 \cdot 69 \rfloor + 1 = 35$. Both cutoffs are fitted separately for each map and interface, and on test data we report the achieved R and C together with alarm rates on all four variants.

Because some systems instead present a fixed number of compartments, we also report the fraction of faults whose

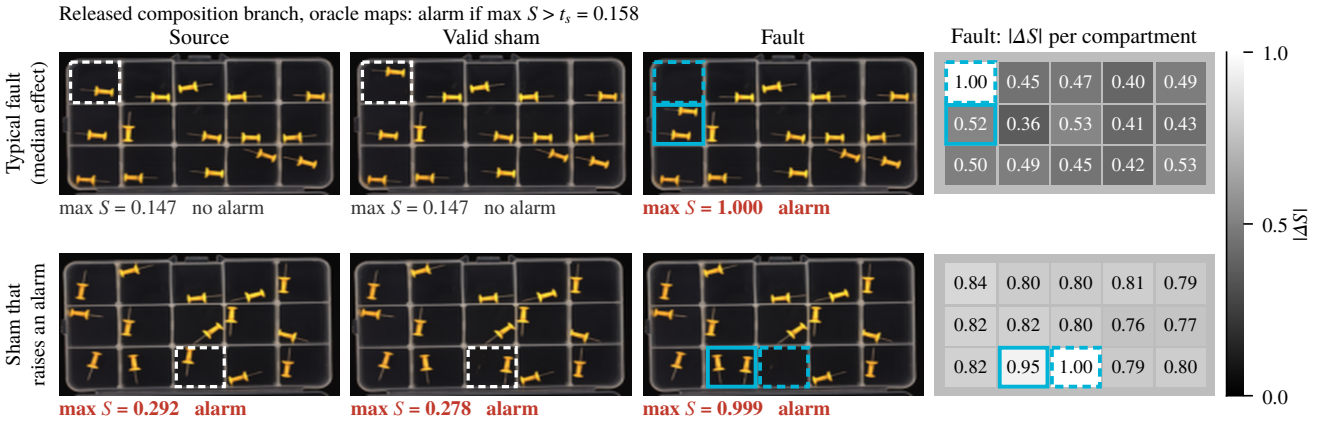


Figure 3. **Adding a pin changes the released map across the whole tray** (real matched edits, oracle maps). Top: the photo whose released fault-minus-sham score difference is the median. Bottom: the first sham that raises an alarm; its source does too. White dashed: compartment of the moved pin; cyan: fault targets. Right: per-compartment change from source to fault, $|S^{\text{fault}} - S^{\text{source}}|$; in the bottom row every compartment changes. Photos: MVTec LOCO AD, CC BY-NC-SA 4.0.

Algorithm 1 Calibrated inspection queue

Require: map model f; cells $\mathcal{E}$; normal validation photos $\mathcal{V}$; development faults $\mathcal{D}$ with targets $\mathcal{T}_j$

$t_s \leftarrow$ 63rd smallest of $\{\max_p f(m_v)(p)\}_{v\in\mathcal{V}}$ ▷ $|\mathcal{V}| = 69$

$G \leftarrow \{g_j(e) : j \in \mathcal{D},\ e \in \mathcal{T}_j\}$, with $g_j(e) = 0$ if j has no alarm

$t_c \leftarrow$ the $\lceil 0.95\,|G| \rceil$-th largest value of G ▷ 95% cutoff

for each test photo i **do**

  $S_i \leftarrow f(m_i)$; **if** $\max_p S_i(p) \le t_s$ **then** $\mathcal{Q}_i \leftarrow \emptyset$

  else $\mathcal{Q}_i \leftarrow \{e \in \mathcal{E} : \max S_i|_e \ge t_c\}$

return R and C from Eq. (3)

two highest-scoring compartments are both targets (*top 2*) and the mean rank of the second target, breaking ties by compartment index. Finally, as a counterpart to sPRO [5, 6], we sweep a common threshold over all compartments, treating targets as positives and untouched fault compartments together with all source and sham compartments as negatives, and integrate target recall up to a false-compartment rate of 0.05, normalized by 0.05 (*region area*). Unlike (R, C), this summary is computed on test data and ignores the scene alarm, so it measures ranking quality rather than the cost at an operating point.

3.4. Footprint supervision

SALAD trains its composition branch on synthetic edits of normal part maps. A discriminative U-Net predicts logits z_p and is fitted to a binary target mask y with

$$\mathcal{L} = 5 \cdot \frac{1}{|P|} \sum_p \mathrm{FL}(z_p, y_p) + \frac{1}{|P|} \sum_p \left|\sigma(z_p) - y_p\right|, \quad (4)$$

where FL is the sigmoid focal loss with $\alpha = 0.25$ and $\gamma = 2$ [12] and the weight 5 is that of the released implementation, σ is the logistic function, and P is the set of pixels. When a part of class k is inserted, producing the edited map m', the released implementation marks every pixel of that class [6],

$$y^{\text{cw}}(p) = \mathbb{1}[m'(p) = k] \qquad \text{(class-wide)}, \quad (5)$$

reflecting the view that an extra screw in a bag may be any of the screws. In a tray, however, only one compartment is over-full, which motivates marking only the footprint F of the pasted donor,

$$y^{\text{fp}}(p) = \mathbb{1}[p \in F] \qquad \text{(footprint)}. \quad (6)$$

For relocations, both policies additionally mark the vacated footprint, shams always receive $y \equiv 0$, and same-image relabelling targets are left unchanged, so the two policies differ only in the insertion part of the mask. In seed 1, a relocation marks on average 3,823 pixels under the class-wide policy and 461 under the footprint policy. The rule is used only to construct training edits; at test time the model observes a part map alone.

3.5. Evaluation protocol

The primary comparison is the difference in cost C between the footprint and class-wide policies on extracted maps. Before testing, we specified that a useful reduction must remove at least one untouched compartment with a 95% interval excluding zero, while keeping the achieved recall at or above 90% and within 5 points of the control, the fault and vacancy alarm rates at or above 95% and within 3 points, and the increase in sham alarms below 5 points. Attributing the effect to the target additionally requires a competent class-wide control on development data, with alarm rates and compartment AUROC of at least 90%. Intervals are computed from

Table 1. **Pushpin inspection on extracted maps** (138 test photos). Every row raises an alarm on all 138 faults and 138 vacancies. *Sham*: valid shams with an alarm. *Top 2*: both targets in the two highest compartments. *2nd*: mean rank of the second target. R and C use the 95% development cutoff (Eq. (3)). *Area*: region area at false-compartment rate ≤ 0.05, in percent. Oracle maps are in the supplement.

Map, compartment score	Sham	Top 2	2nd	R (%)	C	Area
Released, max	7	77.5	2.43	95.7	11.07	81.6
Released, 99th pct.	7	87.0	2.32	96.0	11.41	77.3
Class-wide, seed 1	19	0.0	15.00	97.1	13.00	39.3
Footprint, seed 1	9	87.7	2.13	95.3	**0.20**	**96.3**
Class-wide, seed 2	22	0.0	15.00	96.4	13.00	39.4
Footprint, seed 2	20	74.6	2.28	96.0	**0.45**	89.6

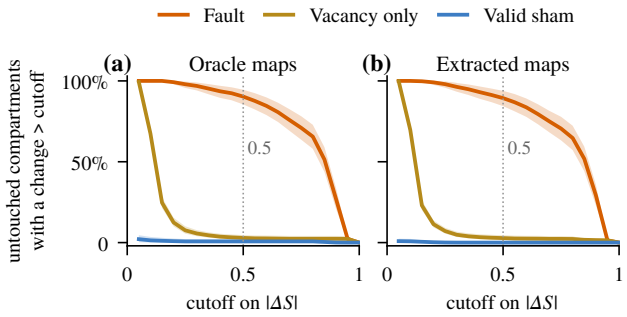


Figure 4. **An extra pin, unlike a vacancy or a valid move, changes the released map across the tray.** Share of the 13 untouched compartments with a pixel whose score changes by more than the cutoff, with 95% intervals over photos.

4,000 bootstrap resamples of the original photographs, in which all variants, policies and interfaces of a photograph are resampled together, and are conditional on the fitted models and cutoffs. Two seeds and two fixed halves of the test photographs (63 and 75) assess stability.

4. Experiments

Implementation details. Each fine-tune starts from the released SALAD composition weights with the reconstruction autoencoder frozen. A batch contains two clean maps, one SALAD-style synthetic edit and one rule-aware edit (a relocation, sham or vacancy) built from the official training photographs. We train for 800 Adam steps and keep the final checkpoint, using two seeds. Within a seed, the two policies receive identical inputs, edits, optimization steps, loss weights and pixel weights, and differ only in y. Further details are given in the supplementary material.

4.1. Diagnosis of the released model

The released composition branch raises an alarm on all 138 faults and all 138 vacancies under both interfaces, and scores every fault above its matched sham (138/138; exact 95% interval 97.4–100%). On oracle maps, the AUROC of the affected compartments rises from 48.6% on the source, which is chance level, to 98.2% after the fault. The ranking is also accurate: on extracted maps, both targets are among the two highest-scoring compartments for 77.5% of faults and among the three highest for 94.2%, and the second target appears at rank 2.43 on average. Scoring compartments by their 99th percentile instead of their maximum raises the top-2 rate to 87.0% (+9.4 percentage points, pp, 95% interval [4.3, 14.5]) with the same alarm and weights.

Good ranking does not, however, translate into a short queue. At the 95% development cutoff, the released map recovers 95.7% of the target compartments but queues 11.07 of the 13 untouched compartments per fault (Tab. 1), so that the inspector examines nearly the entire tray. The 99th-percentile readout, although better at rank two, is no better here (11.41 at 96.0%), and the 98% cutoff raises the cost to 12.56. Conversely, when the cost is held to 0.30 compartments by the fixed-cost cutoff, the released map recovers only 81.9% of the targets (Fig. 5d). A readout that is preferable for a fixed list length can therefore be inferior at a fixed recall.

The reason is that the fault alters the released map across almost the entire tray (Fig. 4). On oracle maps, 90.3% of untouched compartments contain a pixel whose score changes by at least 0.5, although only 5.4% of their pixels change by that amount, indicating sharp, isolated peaks rather than broad regions. The vacancy-only edit, which leaves the same empty compartment, affects only 2.9% of the untouched compartments and the sham 0.7%. The fault-minus-vacancy difference is 87.4 points [82.4, 91.9] on oracle maps and 86.6 [81.4, 91.2] after extraction. The additional pin rather than the vacancy thus causes the widespread response, consistent with a class-wide target that labels every pin as anomalous once a pin is inserted.

4.2. Effect of footprint supervision

Fine-tuning with footprint targets reduces the cost at 95% recall from 11.07 untouched compartments to 0.20 and 0.45 in the two seeds, at an achieved recall of 95.3% and 96.0% (Tab. 1). The paired reductions relative to the released model are 10.88 [10.12, 11.56] and 10.62 [9.86, 11.31] compartments, and relative to the class-wide fine-tunes, the prespecified control, 12.80 [12.72, 12.87] and 12.55 [12.45, 12.65] (Fig. 6). Every fault and vacancy still raises an alarm, and all recall differences have intervals that include zero and remain above the 5-point floor. The reduction is also observed on oracle maps (10.30 and 10.04 relative to the released model) and on both fixed halves of the photographs (-12.87 and -12.75 for seed 1, -12.63 and -12.48 for seed 2). Figure 5 illustrates a typical case in which the re-

leased map queues all 13 untouched compartments and the footprint map queues none.

The advantage holds at the other operating points. At the fixed-cost cutoff, the footprint heads recover 100.0% and 98.2% of targets at achieved costs of 0.38 and 0.57, compared with 81.9% at 0.30 for the released model. In a descriptive sweep at an equal realized cost of 0.50 compartments, recall is 82.6% for the released model and 100.0% and 97.1% for the footprint heads, corresponding to gains of $+17.4$ [13.4, 21.0] and $+14.5$ [9.8, 18.8] points. The region area increases from 81.6% to 96.3% and 89.6%; this ranking-based summary agrees in direction but does not reveal how much shorter the calibrated queue becomes.

Two secondary outcomes depend on the seed. Sham alarms increase from 7 for the released model to 9 for seed 1 ($+1.4$ points [-3.6, 7.2]) but to 20 for seed 2 ($+9.4$ [2.2, 16.7]); relative to the class-wide control, which is the pre-specified comparison, neither seed exceeds the 5-point margin (-7.2 and -1.4). The top-2 rate rises to 87.7% for seed 1 but falls to 74.6% for seed 2, below the released model.

The class-wide fine-tunes, in contrast, retain every alarm but assign high scores to the entire tray: all 13 untouched compartments are queued and the top-2 rate is zero. With a development compartment AUROC of about 48%, they fail the competence criterion of Sec. 3.5. Within this recipe, the comparison with the class-wide arm attributes the difference to the target alone, whereas the comparison with the released model measures the benefit of the complete adaptation, which also introduces rule-aware edits.

4.3. Continuation of the original recipe

To separate the effect of the target from that of the rule-aware edits, we also continue training with SALAD's original synthetic edit families only, replacing the rule-aware slot with a clean map. The class-wide head of seed 2 passes the competence criterion (97.3% development AUROC) and reproduces the behaviour of the released model, with 88.8% of untouched compartments changing by at least 0.5. Footprint targets reduce this fraction to 1.3% and the cost from 11.50 to 7.07, a paired reduction of 4.43 [3.72, 5.14] at 96.0% and 95.3% recall. Seed 1 shows a similar reduction (4.28), but its control does not reach the AUROC criterion. Footprint targets therefore shorten the queue even without rule-aware training, while the rule-aware edits yield the lowest costs.

5. Branch selection under two rules

SALAD fuses three branches: a local RGB appearance map, the composition map studied above, and a global RGB score that pools student features over the scene and over regions defined by the part map [6]. Our replay of the complete released system reproduces the published 100.0% logical AUROC on 91 native logical pushpin faults and 138 normal photographs.

On the matched pushpins, however, the shams reveal a cost that this AUROC does not capture (Fig. 7b). Fusion ranks every fault above its sham but raises an alarm on 130 of 138 extracted shams and 75 of 138 sources, with 55 shams alarming although their source does not. The local RGB branch alone alarms on 128 shams and only 9 sources; since both arms contain the same compositing, it responds to the edit rather than to the rule. The composition branch alarms on 7 shams, and an explicit one-pin-per-compartment rule applied to the same extracted maps raises no sham alarm while detecting every fault.

A second rule shows that the informative branch depends on the rule. MVTec LOCO requires the icon on the upper label of a juice bottle to match its liquid [5]. We retain 89 of 94 eligible normal test bottles and paste intact icons from a disjoint bank of training photographs into the same label region (Fig. 7a); an icon of the same type yields a valid sham and an icon of another type a fault. Both use the same offset, blending and footprint, and their mean blended RGB changes differ by at most half a grey level. Since the liquid, the bottle and the part map are unchanged, the composition branch returns identical maps for all three variants. The global RGB branch ranks all 89 faults above their shams and alarms on every fault but only 14 shams, a fault-minus-sham alarm gap of $+84.3$ points [76.4, 91.0] (Fig. 7d), slightly above that of fusion ($+79.8$ [70.8, 87.6]). The local RGB branch ranks the upper label first among five bottle regions for 74.2% of faults but also for 85.4% of shams, and alarms on 25 faults and 28 shams. The two rules therefore favour different branches: composition evidence decides placement in a tray, whereas global appearance captures a label–liquid correspondence that the part map cannot express.

6. Response evidence in part graphs

The matched design also applies to graphs of parts. We build gland graphs from GlaS histology images [17] and cardiac structure graphs from HVSMR-2.0 [14], and apply *degree-preserving switches* that remove two links and add two others so that every node keeps its degree. Readouts are evaluated by how well they rank the four switched pairs. On GlaS, adding local gland spacing to typed centroid distance improves the edited-graph AUROC by 15.3 points [10.9, 20.1], and the gain persists for additional length-matched switches and under joint matching of raw and local distances ($+16.8$ [11.3, 23.0]); these tiers reuse donors and involve no pathological labels.

For a candidate link $e = (u, v)$ between nodes u and v, let z_e be the predicted logit and $a_e \in \{0, 1\}$ the observed link state. For a link predictor, the Bernoulli loss $\mathrm{softplus}(z_e) - a_e z_e$ changes with a_e even when the evidence z_e is fixed. A pair MLP observes only geometry and roles, so its probability cannot change and it ranks by response at chance level (Fig. 8b). A query-masked neighbourhood

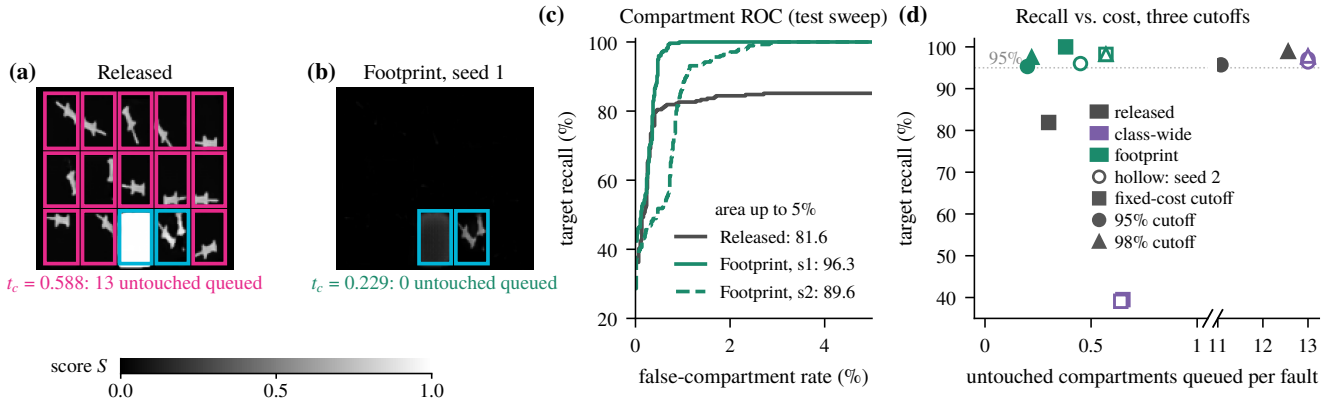


Figure 5. **Footprint supervision empties the queue.** (a,b) Extracted maps of the median fault (greyscale); cyan: targets, pink: untouched compartments at or above the pre-specified 95% cutoff t_c. (c) Compartment ROC from a test sweep, with the area up to 5% from Tab. 1 (class-wide curves, area 39.3–39.4, omitted). (d) Recall against cost at the fixed-cost (square), 95% (circle) and 98% (triangle) development cutoffs; hollow: seed 2; the axis is broken between 1 and 11.

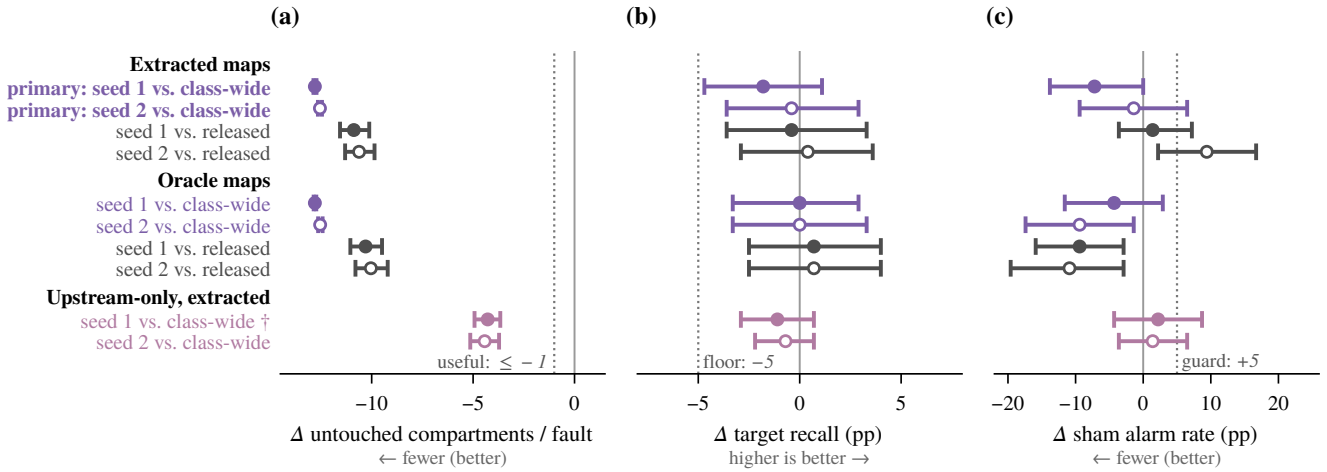


Figure 6. **Footprint targets remove 10 to 13 untouched compartments per fault without losing recall.** Paired differences at the pre-specified cutoffs, footprint minus each comparator, with 95% intervals; hollow: seed 2. Bold: the pre-specified primary contrast. Dotted: pre-specified thresholds. †: this seed's control fails the competence check (Sec. 4.3). Values: supplement Tabs. 3 and 8.

model has an additional route: removing the queried link before computing the degree d_u of an endpoint yields $d_u - a_e$, so a switch changes the masked degree at the four target pairs although both graphs share the same degree sequence, an effect related to the target-link exclusion studied in [23]. Ranked by absolute change in probability, the masked GNN recovers 87.5% of the switched pairs among its top four (Fig. 8c). Holding its degree inputs fixed reduces this to 33.7%, and refitting with mean neighbourhood aggregation and no degree inputs on the same four GlaS folds, which reaches 95.7–98.2% development AUROC on normal links, reduces it to 46.6%, 40.9 points lower [32.8, 49.8]. Since mean aggregation still admits indirect cues such as an empty neighbourhood, both interventions indicate a degree pathway, and recovery of known graph edits should not be interpreted as evidence of validity.

7. Conclusion

We have shown that matched valid and invalid edits turn a logical anomaly benchmark into a measurement of inspection effort. A detector that detects every fault and ranks its targets well can nevertheless direct the inspector to 11 of 13 untouched compartments at high recall, a cost that image AUROC, top-k rates and region overlap do not reveal but the recall–cost pair at a development cutoff does. This cost is largely determined by the training target: marking every part of the inserted class is appropriate for interchangeable parts, but for parts tied to locations it teaches the network that a single extra pin renders the whole tray anomalous. Marking

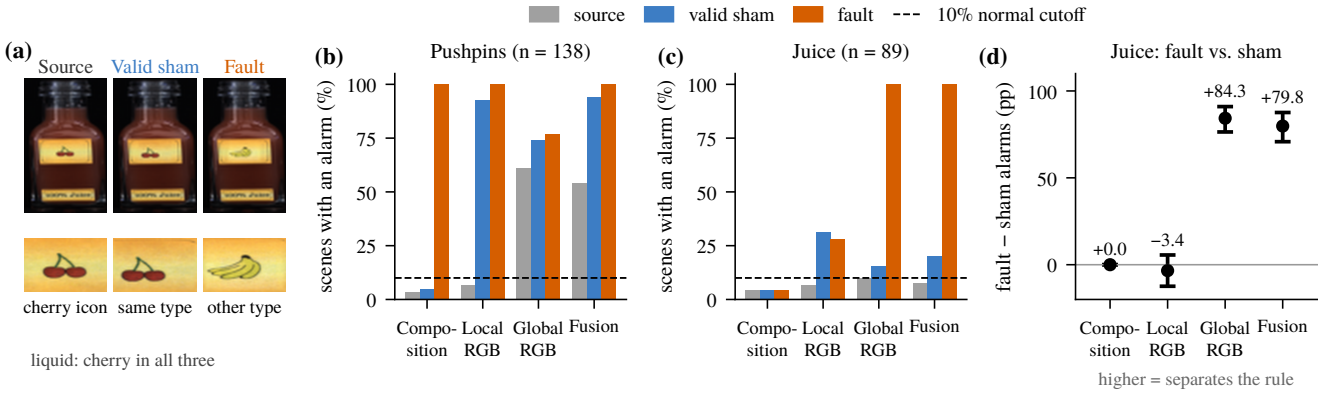


Figure 7. **Which SALAD branch separates a fault from a valid edit?** (a) Juice edit: a same-type (valid) or other-type (fault) icon, pasted identically. (b,c) Scenes alarmed per branch at its own 10% normal cutoff (dashed); on pushpin composites, global RGB and fusion also alarm on most untouched sources. (d) Juice fault-minus-sham alarm gap, 95% intervals. Photos: MVTec LOCO AD, CC BY-NC-SA 4.0.

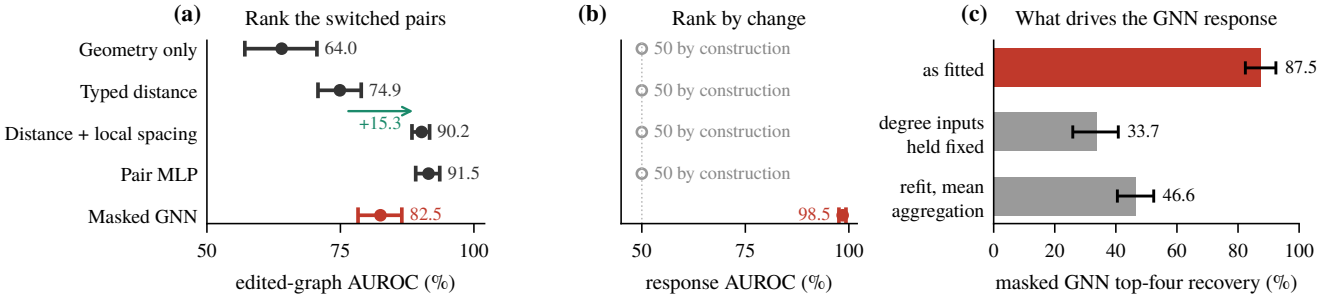


Figure 8. **A masked GNN responds to switched links through node degree** (78 GlaS fields, 12 donors). (a) Edited-graph AUROC; local spacing adds 15.3 points to typed distance. (b) AUROC of the change between matched graphs; hollow: 50 by construction. (c) Top-four recovery of the masked GNN as fitted, with degree inputs fixed, and refit with mean aggregation. Donor-paired 95% intervals.

only the pasted part reduces the cost to below half a compartment. The same matched design identifies the branch that carries the evidence under a new rule and exposes a degree pathway in a masked graph model. We argue that logical anomaly detectors should be evaluated by recall together with inspection cost against matched valid edits.

The evidence has a clear scope. Our strongest result concerns two compartments in a registered tray. Both seeds improve the calibrated high-recall queue, whereas the top-2 ranking and sham specificity depend on the seed. Training uses edits constructed from a known rule, so the gain is established for that setting. The interventions are synthetic composites of held-out photographs that match the changed pixels, the cutout and the displacement, but they are not physical rearrangements, and physical tray identities are unavailable. All intervals are conditional on the fitted heads and cutoffs. Because the class-wide rule-aware controls fail the competence criterion, mechanistic claims rest on the continuation of the original recipe, for which one seed passes. The juice experiment covers one additional rule and the graph analysis one masking design; native LOCO results and an exploratory cardiac query are reported in the supplementary material. Three next steps follow directly from these findings: physical rearrangements of real trays to confirm the effect outside composites, a retraining of the full released recipe with footprint targets from scratch, and the same recall–cost evaluation on further logical rules where parts are tied to locations.

LLM Usage Statement

OpenAI Codex was used to develop and implement the revision experiments, run and check the code, inspect public sources and data provenance, produce figures and tables from saved outputs, and draft and edit the main text and supplementary material. Anthropic Claude was used to restructure and edit the text and to redraw the figures from the recorded outputs. Parts of Figs. 1 and 2 (tray and network illustrations) were generated with BioRender, and all labels were typeset separately; all quantitative panels use deterministic plotting of the recorded results and the documented photograph composites. The authors take full responsibility for the content.

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